THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **The Wonderful World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. To find whether the volume of water affects its boiling time, a student boils 100 mL in a steel pot on a strong flame and 200 mL in a glass beaker on a weak flame. Why is this test unfair? 1

(a) the volumes of water are different
(b) the water is heated at all
(c) time is measured with a clock
(d) the container and the flame also differ along with the volume

2. A study found that more ice cream is sold when more people get sunburnt. The best explanation is: 1

(a) both depend on hot weather **(b)** ice cream causes sunburn
(c) sunburn causes ice cream sales **(d)** it is magic

3. Suppose two groups test the same hypothesis and get opposite results. What should be done first? 1

(a) toss a coin **(b)** compare the methods and repeat carefully
(c) ignore both **(d)** believe one group

4. In an experiment on how salt affects the boiling point, two things were changed: salt amount and heater power. The error is: 1

(a) more than one variable was changed **(b)** water is cold
(c) too many readings **(d)** salt is a solid

5. Plants A, B, C are given 100 mL, 200 mL, and 300 mL of water in same soil and light, and A grows least. Which is the changed variable? 1

(a) amount of soil **(b)** type of plant **(c)** amount of light **(d)** amount of water

6. Which of the following is not good practice while recording data? 1

(a) noting the date
(b) rounding off numbers to look neat without noting it
(c) giving a title
(d) writing units

7. A scientist is convinced that her hypothesis is right, but the data shows otherwise. The correct action is to: 1

(a) discard the data **(b)** repeat until the data matches
(c) change the readings **(d)** revise the hypothesis

8. In the above experiment, which variable is measured as the outcome? 1

(a) growth of the plant **(b)** amount of water
(c) soil **(d)** sunlight

9. **Assertion (A):** A hypothesis is a guess that can be tested. 1
Reason (R): An experiment can show whether a hypothesis is supported or not.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Observation is the first step of the scientific method. 1
Reason (R): A telescope helps us observe objects that are very far away in the sky.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In an experiment on plant growth, A is kept in sunlight and B in shade, but A is also given more water. Why is this unfair? 2

12. How can a scientific model such as a globe help us? 2

13. Give one example in which technology causes a problem and how science may solve it. 2

14. Why do scientists use a control group? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Design a fair test to compare which of three fabrics dries the fastest. Name controlled variables. 3

16. Explain the role of curiosity, observation and reasoning in the discovery of gravity by Newton. 3

17. Explain the term 'variable' and give examples of three variables in an experiment on dissolving sugar. 3

OR

In a plant experiment, plants P (light, water), Q (dark, water), R (light, no water). State what each comparison tests. 3

18. Compare the use of a thermometer with feeling with the hand to judge temperature. 3

OR

Describe two ways in which a scientific idea may be proved wrong. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Riya noticed that the plant on her balcony grew taller than the plant kept inside her cupboard. She guessed that plants need sunlight to grow well. She took two identical pots with the same soil and gave both the same amount of water. She kept one in sunlight and the other in a dark cupboard, and noted the height of both plants every two days for ten days. 5
(i) What is Riya's guess called?
(ii) Which one thing was different for the two pots?
(iii) Name the step in which she noted the heights every two days.
(iv) Why did she keep the soil and water the same? What is the likely conclusion?

20. Three students measured the time of 20 pendulum swings: 28 s, 30 s, 32 s. Find the average, the time for one swing, and say why 20 swings are timed. 5

OR

Explain how errors can occur in measurement and suggest five ways to reduce them. 5

END OF QUESTION PAPER — **Total: 40 Marks**